

Tandem

Two RMS soft-knee compressors in series on one panel.



Overview

Tandem puts two classic RMS / VCA compressors one after the other. Each stage does a little of the work, which is the old studio trick for heavy compression that still sounds open: the first stage levels the performance gently, the second catches what gets through.

Both stages measure loudness (RMS) rather than peaks, so they react the way your ears do. There are no attack and release knobs - the timing follows the music, as on the original hardware: the further the signal goes over the threshold, the faster the stage reacts, and it recovers at a steady pace.

Reach for it for

- Vocals that need to sit still in a mix
- Drum and mix buses
- Bass guitar
- Ducking a music bed under a voice (EXT KEY)
- Creative pumping and reverse-dynamics effects (negative ratios)

Controls

Stage 1 and Stage 2

The two stages have the same main controls. The signal goes through Stage 1, then Stage 2.

IN	Switches the stage in or out. With a stage out, the audio passes straight through it.
THRESHOLD (red cap)	The level where the stage starts working, from -60 to 0 dB. Turn it down (left) for more compression. Because the detector measures loudness, the threshold relates to the average level rather than the peaks.
RATIO (yellow cap)	How strongly the stage compresses once the level is over the threshold. 1:1 does nothing; 2:1 and 4:1 are moderate; INF:1 (marked in red) is limiting - nothing gets louder. Past INF the ratio becomes negative (-2:1, -1:1): the louder the input, the quieter the output. That reverses the dynamics, for pumping and swelling effects.
OUTPUT (blue cap)	The stage's make-up gain, from -20 to +20 dB. Use it to bring the level back up after compression.
BELOW / KNEE / ABOVE	Three lights showing where the level sits: green below the threshold, amber in the soft knee, red above it.
GR meter	The row of 12 lights under the knobs shows the gain reduction of that stage, from 1 to 30 dB.
HARD	Off (default): a soft knee - the ratio eases in gradually over 10 dB around the threshold, which is smoother and more transparent. On: the full ratio starts right at the threshold, for a firmer, more obvious effect.

Stage 1 only

DECOUPLE	Off: both sides of the stereo signal get exactly the same gain, so the stereo image stays still. On: each side mostly follows its own level (80% its own, 20% shared), like unlinking a bus compressor. The sound becomes wider and livelier, but the image can move a little.
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EXT KEY	Stage 1 listens to the key input (plug-in pins 3 and 4) instead of the audio it is compressing. Use it for ducking: route a voice or kick to pins 3/4 and the music dips whenever it plays. Stage 2 always listens to Stage 1's output.
KEY IN	Lights when a signal arrives on the key input - a quick check that your routing works.

Stage 2 only

GRIT	Drives Stage 2's VCA into gentle 2nd and 3rd harmonic distortion. The harder Stage 2 works, the grittier it gets, so it adds attitude on the loud parts and stays clean on the quiet ones.
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Master

Transfer curve	Shows the input-to-output curve of both stages together. The white dot rides on it with the live level, so you can see exactly where the music sits.
SC HPF	A high-pass filter on what both stages listen to (fully left = off, up to 400 Hz). Raise it so the bass doesn't make the whole mix pump; the bass itself isn't filtered.
MIX	Parallel compression: blends the dry signal back in. 100% is fully compressed; lower settings keep the transients of the dry signal under the compressed sound.
IN / OUT meters, TOTAL GR	Input and output level, and the combined gain reduction of both stages.

The header

PRESETS	Opens the list of factory presets. Choosing one sets the plugin's controls for that job; the input and output stages are left as they are. The last entry resets everything to the defaults.
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S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
OUTPUT	Folds the input and output stages out below the panel (and back in). The light shows when they are open.

Input and output stages (**OUTPUT fold-out**)

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The tag shows how much it is working.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
BIAS	Shapes the saturation: more bias adds more even (2nd) harmonics, for a warmer, rounder colour.
LO BAND / HI BAND	Where the low and high saturation bands start (40 - 400 Hz and 2 - 16 kHz).
CEILING / THRESH	With CLIP: the level the peaks are softly clipped at. With LIMIT: the threshold of a brickwall limiter that holds the peaks at -0.3 dBFS - pulling it down limits harder and makes the output louder.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER).

Presets

Gentle Glue	Light two-stage smoothing for a mix or group bus.
Vocal Leveller	Stage 1 evens the performance, Stage 2 catches the peaks; a little SC HPF.
Vocal Upfront	Heavier vocal control with a hard knee and GRIT on Stage 2.
Drum Bus Stack	Stage 1 decoupled for a lively image, GRIT on Stage 2, 70% mix.
Parallel Crush	Stage 2 limiting hard with GRIT, blended in at 35%.
Bass Even	Steady bass level with a little GRIT.
Acoustic Guitar	Gentle, with Stage 1 decoupled for width.
Peak Catcher	Light levelling, then a hard-knee limit at -6 dB.
Negative Pump	Stage 2 past INF for rhythmic pumping, 60% mix.
Reverse Swell	Stage 2 at -1:1: loud hits dip and the tails swell up; 50% mix.
Ducker (ext key)	Stage 1 ducks from the key input - route the voice or kick to pins 3/4.
Reset all to defaults	Every control back to its starting value.

Using the key input (EXT KEY)

1. On the track with Tandem, open ROUTE and set the track to 4 channels.
2. On the track you want to duck from (e.g. the voice), add a send to Tandem's track and set its destination to channels 3/4.
3. Switch EXT KEY on. KEY IN lights when the key signal arrives; lower Stage 1's THRESHOLD until the music dips by the amount you want.

Tips

- Start with Stage 2's ratio at INF and its threshold just under your peaks: it becomes a safety limiter after Stage 1's gentle levelling.
- If the low end makes the whole mix duck, raise SC HPF to 80-120 Hz.
- Negative ratios sound best on sustained material (pads, rooms, reverb returns) and with MIX below 100%.
- Use the transfer curve: if the white dot sits far up the flat part of the curve, you're compressing more than you probably need.

File: gfx-tandem.jsfx